1. Clinical Question & Core Hypothesis

Clinical Question

In muscle-invasive disease treated with pembrolizumab plus chemotherapy, why do some tumors with high pre-treatment TIL density fail to achieve clinical response? The single path links chemotherapy-induced cell death, KRAS signaling, and therapy-associated mutagenesis, suggesting KRAS status may modulate the immunogenic consequences of cytotoxic therapy and PD-1 blockade. 2) Tier 1 path used: the only presented path (ClinicalTrial → pembrolizumab+chemotherapy → chemotherapy induces cell death → cell death ↔ KRAS → drug (cisplatin) induces genetic mutations). I built on the mechanistic backbone that KRAS signaling can alter cell-death programs and that chemotherapy can both induce immunogenic cell death and cause mutagenesis. 3) Tier 2 evidence: there are no additional tiers provided; supporting plausibility comes from known biology that KRAS activation can dampen pro-immunogenic apoptosis/necrosis and that platinum agents (e.g., cisplatin) cause DNA damage that can create or select for resistant clones. These support a model where KRAS alters the balance between immunogenic cell death and therapy-driven evolution. 4) Why therapy_response endpoint: the path centers on a clinical trial of pembrolizumab plus chemotherapy—hence response to this combined therapy is the most directly relevant and actionable endpoint. 5) Deliberate exclusions: I excluded placing cisplatin-induced mutagenesis or generic DNA-repair genes into the core variables to keep the hypothesis focused; those processes (chemotherapy-induced mutagenesis, immunogenic cell death programs, emergent resistant clones) are noted as mechanistic modifiers in the notes but not forced into the primary hypothesis to satisfy the Focusing Rule?

Working Hypothesis

Activating KRAS mutations modulate tumor-intrinsic immunogenicity such that high baseline TIL density does not predict response to pembrolizumab plus chemotherapy in muscle-invasive tumors. KRAS-driven signaling attenuates immunogenic cell death and promotes immune evasion under cytotoxic therapy, lowering response rates in KRAS-mutant, TIL-high tumors compared with KRAS-wildtype, TIL-high tumors.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	KRAS
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

KRAS: KRAS signaling, and therapy-associated mutagenesis, suggesting KRAS status may modulate the immunogenic consequences of cytotoxic therapy and PD-1 blockade.

3. Clinically Testable Predictions

Prediction 1: perform retrospective analyses in trials of pembrolizumab+chemotherapy with known KRAS status and TIL density, testing for an interaction between KRAS mutation and TIL density on therapy_response;

Prediction 2: profile pre-treatment tumors (IHC for TILs, PD-L1; RNA-seq for immune signatures) in KRAS-mutant vs KRAS-wildtype groups;

Prediction 3: in vitro assays comparing KRAS-mutant and wild-type cell lines for chemotherapy-induced ICD markers (calreticulin exposure, HMGB1, ATP release) and subsequent T-cell–mediated cytotoxicity;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions